

Harun B | Machine Learning Engineer

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SUMMARY

Machine Learning Engineer with nearly **5 years of experience** building and deploying production ML systems (recommendation, search/NLP, forecasting) on AWS SageMaker. Strong in Python, SQL, PyTorch, and MLOps (MLflow, Airflow, Docker, model monitoring), with LLM and multi-agent (LangGraph) project experience. Delivered a 17% lift in purchase conversion and a 23% improvement in forecast accuracy.

TECHNICAL SKILLS

Programming & Data: Python, SQL, PySpark, Pandas, NumPy, Git/GitHub, Matplotlib, Seaborn
ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, Distributed Training, Classification, Regression, Clustering (K-Means, DBSCAN), Time-Series Forecasting (RNN/LSTM), Computer Vision (CNN), Recommendation Systems, Feature Engineering, A/B Testing
Generative AI & NLP: Large Language Models (LLMs: OpenAI, Claude, Gemini), Retrieval-Augmented Generation (RAG), Transformers (GPT, BERT, HuggingFace), Semantic Search, Vector Databases (Pinecone, ChromaDB), LlamaIndex, Fine-Tuning (LoRA/QLoRA), Prompt Engineering
Agentic AI: LangGraph, LangChain, CrewAI, Model Context Protocol (MCP), Multi-Agent Systems, Agent Orchestration, ReAct, Tool/Function Calling, Agent Memory
MLOps & LLMops: MLflow, Weights & Biases, LangSmith, Agent Evals, Tracing & Observability, Model Monitoring & Drift Detection, Feature Store, CI/CD, Docker, Kubernetes, Airflow, Prefect, FastAPI, Flask, REST APIs
Cloud & Inference: AWS (S3, SageMaker, Bedrock, Lambda), GCP, Azure AI, vLLM, Triton Inference Server, ONNX Runtime, TensorFlow Lite
Explainability & Ethics: SHAP, LIME, Bias Detection

PROFESSIONAL EXPERIENCE

Etsy, NY, USA Jan 2025 - Current

Machine Learning Engineer

- Heightened purchase conversions 17% by deploying recommendation models on AWS SageMaker integrated into live customer personalization systems.
- Reduced bounce rates 14% by applying Transformer-based NLP models (BERT, GPT) to improve search accuracy and discovery relevance.
- Decreased deployment cycle time 40% through end-to-end automation of ML pipelines using Airflow, MLflow, and Docker orchestration.
- Enhanced transparency of deployed models by integrating SHAP and LIME explainability, strengthening trust and compliance across regulated environments.
- Improved real-time inference speed 28% by optimizing scalable APIs, enabling faster predictions during peak traffic across ecommerce platforms.
- Delivered measurable 11% uplift in user engagement by designing and validating A/B tests for search and recommendation engines.
- Maintained predictive performance above 90% accuracy by monitoring drift and retraining models using automated MLflow pipelines.
- Accelerated ML adoption by integrating engineered models directly into production applications used daily by cross-functional business stakeholders.

Fractal Analytics, India Mar 2020 - Jun 2023

Machine Learning Engineer

- Delivered 92%+ model accuracy by developing regression, classification, and clustering solutions for large-scale healthcare and retail datasets.
- Amplified forecast precision 23% by designing CNN-based image analytics and LSTM time-series models for enterprise supply chain optimization.
- Reduced latency 35% by containerizing ML APIs with Flask, FastAPI, and Docker for scalable production implementations.
- Streamlined workflows by automating preprocessing, training, and retraining tasks with Airflow and Prefect, reducing manual intervention significantly.
- Improved executive decision-making by building Tableau and Power BI dashboards translating predictive outputs into actionable KPIs.
- Boosted model robustness by engineering domain-specific features across structured and unstructured datasets for diverse client projects.
- Validated predictive reliability by applying metrics such as precision, recall, F1, and ROC-AUC across all deployed models.
- Strengthened client outcomes by delivering complete ML solutions collaboratively with engineers, analysts, and business stakeholders across industries.

EDUCATION

M.S. in Information Systems and Technology Aug 2023 - May 2025

Wilmington University, New Castle, DE

Bachelor of Technology in Computer Science and Engineering Jul 2017 - Jun 2021

Malla Reddy Institute of Engineering and Technology, India

CERTIFICATION

Machine Learning Specialization - Coursera (Andrew Ng, DeepLearning.AI)

AWS Certified Cloud Practitioner

PROJECTS

Multi-Agent Research & Report Mar 2026 – Jun 2026

- Built a multi-agent workflow in LangGraph with planner, researcher, and writer agents that turns a topic and report outline into a structured, source-cited research report.
- Implemented shared graph state, conditional routing, and tool calling so agents generate search queries, retrieve web sources, and draft each section from retrieved evidence.
- Traced and debugged agent runs with LangSmith to inspect tool calls, latency, and token usage.

Build GPT from Scratch Jul 2025 - Aug 2025

- Implemented a decoder-only Transformer (multi-head self-attention, positional embeddings, residual connections, layer norm) and trained a character-level language model on tiny shakespeare dataset
- Improved training efficiency and accuracy by applying mini-batching, cross-entropy loss, backpropagation, and mixed-precision optimization during model training.